

RAJALAKSHMI
ENGINEERING COLLEGE
An AUTONOMOUS Institution
Affiliated to ANNA UNIVERSITY, Chennai

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AI POWERED INCIDENT RESPONSE SYSTEM

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ABSTRACT

The increasing number of cyber threats has made manual incident response slow and inefficient. This project proposes an **AI-Powered Incident Response System** that automates the detection, analysis, prioritization, and response to cybersecurity incidents using Artificial Intelligence. The system employs **Intelligent Agents** to monitor security events and a **Rule-Based Expert System** with **Knowledge Representation** to classify incidents. **Forward Chaining** is used to recommend appropriate response actions, while **Heuristic Search** prioritizes incidents based on their severity and impact. An interactive dashboard displays incident details, severity levels, and recommended mitigation strategies, enabling faster decision-making and improving the overall efficiency of cybersecurity incident management.

OBJECTIVE

The primary objective of this project is to develop an **AI-Powered Incident Response System** that automates the detection, classification, prioritization, and response to cybersecurity incidents. The system utilizes **Intelligent Agents**, **Knowledge Representation**, **Forward Chaining**, and **Heuristic Search** to analyze security events and recommend appropriate mitigation actions. It also provides an interactive dashboard for real-time incident monitoring, enabling faster decision-making, reducing manual effort, and improving the efficiency of cybersecurity incident management.

Literature Survey

TITLE	YEAR	AUTHOR	TECHNIQUE
AI-Driven Cybersecurity Incident Response: A Survey	2023	S. Sharma, R. Gupta	Machine Learning, Rule-Based Systems, Intelligent Agents
Artificial Intelligence for Cybersecurity: A Systematic Literature Review	2022	A. Ahmad, J. Khan	Knowledge Representation, Expert Systems, Threat Detection
Automated Incident Response Using Artificial Intelligence	2021	M. Alshammari, P. Smith	Forward Chaining, Heuristic Search, Rule-Based Expert System

EXISTING SYSTEM

- Security alerts generated by firewalls, IDS, and antivirus systems are analyzed manually by security analysts.
- Existing incident response processes are time-consuming and require continuous human monitoring.
- Traditional systems rely on predefined signatures, making them ineffective against new or unknown threats.
- There is no intelligent mechanism to automatically prioritize incidents based on severity and impact.
- Delayed detection and response can increase the risk of successful cyberattacks and data breaches.
- Existing solutions provide limited automation and lack AI-based decision-making for recommending appropriate response actions.

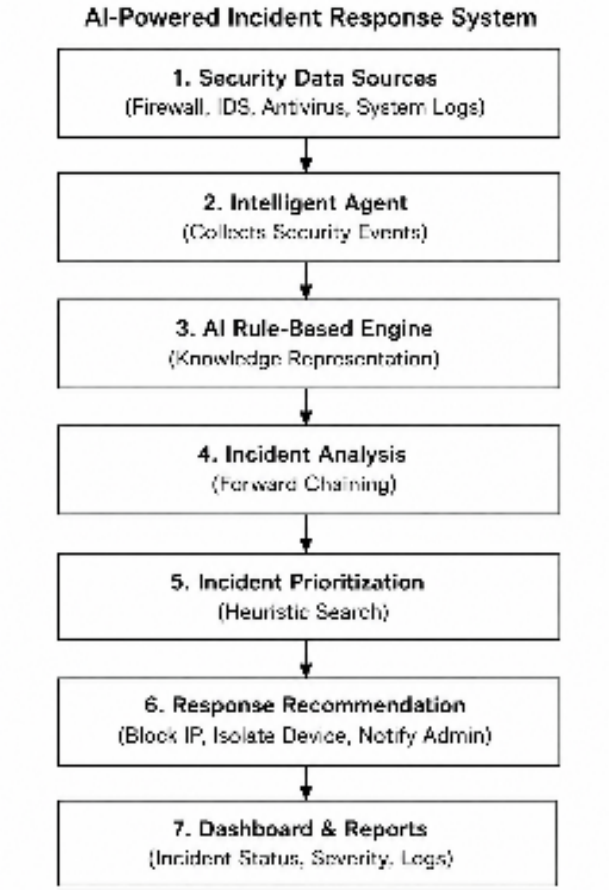
PROPOSED SYSTEM

- Develop an **AI-Powered Incident Response System** to automate the detection and analysis of cybersecurity incidents.
- Use **Intelligent Agents** to continuously monitor security events from multiple sources.
- Implement a **Rule-Based Expert System** with **Knowledge Representation** to classify cyber threats.
- Apply **Forward Chaining** to infer appropriate response actions based on predefined security rules.
- Use **Heuristic Search** to prioritize incidents according to their severity, impact, and risk level.
- Provide an interactive dashboard for real-time monitoring, incident tracking, and recommended mitigation strategies

Novelty in Proposed System

- Integrates **Artificial Intelligence** to automate cybersecurity incident response.
- Uses **Intelligent Agents** for continuous monitoring of security events.
- Employs a **Rule-Based Expert System** with **Forward Chaining** for intelligent decision-making.
- Prioritizes incidents using **Heuristic Search**, enabling faster handling of critical threats.
- Provides **real-time recommendations** for mitigation instead of relying solely on manual analysis.
- Features an **interactive dashboard** for incident visualization, tracking, and response management.

BLOCK DIAGRAM(PROPOSED SYSTEM)



Hardware/Software Requirements

Hardware Requirements	Software Requirements
Processor: Intel Core i3 or above	Operating System: Windows 10/11
RAM: 8 GB Minimum	Frontend: React.js, HTML, CSS, JavaScript
Storage: 256 GB SSD	Backend: Spring Boot (Java)
Internet Connection	Database: MySQL
Laptop/Desktop	AI Tools: Python, Scikit-learn, Experta
	IDE: VS Code / IntelliJ IDEA
	API Testing: Postman

References

1. **Russell, S., & Norvig, P.** *Artificial Intelligence: A Modern Approach*, 4th Edition, Pearson, 2021.
2. **William Stallings.** *Network Security Essentials: Applications and Standards*, 7th Edition, Pearson, 2022.
3. **NIST Special Publication 800-61 Rev.2.** *Computer Security Incident Handling Guide*, National Institute of Standards and Technology.
4. **MITRE ATT&CK Framework.** https://attack.mitre.org/ **OWASP Foundation.** *OWASP Top 10 Web Application Security Risks.* https://owasp.org/